MIL-F-48307(PA)
30 September 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

FUZE, IMPACT, ATOMIC WEAPON: M1110

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of contact fuze entitled Fuze, Impact, Atomic Weapon: M1110 which is used in the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8800854	- Contact.
8808362	- Container, Packing, Ammunition, for Domestic (CONUS) Shipments of Special Weapons Items, Packing and Marking.
8813366	- Contact.
8813373	- Fuze, Impact, Atomic Weapon. M1110.
8822043	- Container, Metal, Ammunition, for Fuze, Impact, Atomic Weapon M1110.
EL-8842562	- Equipment List.

FSC1135

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The impact fuze shall comply with all requirements specified on the drawing (dwg) 8813373, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Resistance.

3.5.1 Circuit resistance, actuated condition. Resistances in actuated condition shall be in accordance with Table I when measured as specified in 4.4.1.1.

Table I

From	To	Resistance
AIJI Pin A	Cover	0.040 ohms maximum (max)
AIJI Pin B	Cover	0.040 ohms max
AIJ2 Pin A	Cover	0.040 ohms max
AIJ2 Pin B	Cover	0.040 ohms max
AIJI Pin C	AIJ2 Pin C	0.040 ohms max

3.5.2 Circuit resistance, non-actuated condition. Resistance in the non-actuated condition shall be in accordance with Table II when measured as specified in 4.4.1.2.

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Table II

From	To	Resistance
AIJI Pin A	AIJI Pin B	0.040 ohms max
AIJI Pin A	AIJ2 Pin B	0.040 ohms max
AIJ2 Pin A	AIJ2 Pin B	0.040 ohms max

3.5.3 Insulation resistance. Resistance in the non-actuated condition shall be in accordance with Table III except the insulation resistance after environmental tests shall be 20 megohms minimum. The test shall be conducted as specified in 4.4.1.3.

Table III

From	To	Resistance
AIJI Pin A	Cover	200 megohms minimum (min) at 500 ± 50 vdc *

*Volts direct current (vdc)

3.6 Environmental testing. The impact fuze shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The impact fuze shall also comply with 1.2 and 4.4.1.3 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.6.1 Shock.

3.6.1.1 Non-actuation shock. The impact fuze shall be tested at the shock amplitudes, temperatures and for the times indicated in Table IV. The impact fuze shall be shocked one time in each of three mutually perpendicular major axes at each temperature specified. The circuit from AIJI Pin A to the cover shall show no evidence of closure when monitored with the circuit of Figure 1, or equivalent, during shocks. The impact fuze shall meet the requirements of 3.5.2 and 3.5.3 after being tested as specified in 4.4.2.1.

Table IV

Peak shock Amplitudes	Beginning to End Half-sine Shock Wave Time Duration	Temperatures
110 ± 10 g's*	10 ± 2 milliseconds	80 ± 20°F, -65 ± 5°F, 160 ± 5°F

*gravity units (g's)

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3.6.1.2 No-load actuation shock. The impact fuze shall be tested at the shock amplitudes, and for the times indicated in Table V. The impact fuze shall be shocked one time in each of three mutually perpendicular major axes at each temperature specified. The circuit AIJI Pin A to the cover shall close during each shock for a minimum continuous period of 200 microseconds. The impact fuze shall meet the requirements of 3.5.2 and 3.5.3 after being tested as specified in 4.4.2.2.

Table V

Peak Shock Amplitude	Beginning to End Half-sine Shock Wave Time Duration	Temperatures
250 ± 20 g's	9 ± 3 milliseconds	80 ± 20°F, -65 ± 5°F, 160 ± 5°F

3.6.1.3 50-ampere load-actuation shock. The impact fuze shall be subjected to one shock only in one axis at the shock amplitude; time duration and while at the temperature specified in Table V. The impact fuze shall carry 50-5, +0 amperes direct current with a maximum voltage drop from AIJI Pin A to the cover of 2.0 volts for a minimum continuous period of 200 microseconds. The test shall be conducted as specified in 4.4.2.3.

3.6.2 Acceleration. The impact fuze shall withstand 100 + 10 g's of acceleration for 50 seconds minimum along each of three mutually perpendicular axes without evidence of damage or deterioration affecting functioning, and the circuit from AIJI Pin A to the cover shall show no evidence of closure when tested as specified in 4.4.3.1. The fuze shall meet the requirements of 3.5.2 and 3.5.3 after this test.

3.6.3 Vibration. The assembly shall be vibrated at the frequencies, accelerations, temperatures, and for the time specified in Table VI without evidence of damage such as cracks, chips, separations or loose parts. The circuit from AIJI Pin A to cover shall show contact closure of no more than 10 microseconds during vibration. The assembly shall comply with the requirements of 3.5.2 and 3.5.3 after being tested as specified in 4.4.3.2.

Table VI

Frequency Range of Test Cycle	Time For* Each Cycle	G Level	Test time per axis (minutes) at temp 80 + 20°F	160 + 5°F	-65 + 5°F
10-20-10 Hz**	60 ± 10 sec	2.0 ± 0.2	30 min	15 min	15 min
21-180-21 Hz	5 ± 1 min	10 ± 1	15 min	15 min	15 min
181-1000-181 Hz	5 ± 1 min	20 ± 0.2	15 min	15 min	15 min

*Sweep time of frequency range from the low value to the high value and back to the low.

**Hertz (Hz)

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3.6.4 Temperature shock. The impact fuze shall withstand exposure to temperature shocks of minus (-) 65 degrees Fahrenheit (°F) plus or minus (±) 5°F to 160 ± 5°F without evidence of damage or deterioration affecting functioning when tested as specified in 4.4.3.3. Upon completion of this test the impact fuze shall meet the requirements of 3.5.2, 3.5.3, 3.6.1.1 and 3.6.1.2.

3.6.5 Humidity. The impact fuze shall withstand exposure to a relative humidity of 95 ± 5 percent at temperatures cycling from 80 ± 20°F to 160 ± 5°F without evidence of damage or deterioration when tested as specified in 4.4.3.4. Upon completion of this test the impact fuze shall meet the requirements of 3.5.2, 3.5.3, 3.6.1.1 and 3.6.1.2.

3.7 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460(MU).

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 9 complete assemblies (see dwg 8813373) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VII. Tests shall be conducted in order specified in Table VII.

Table VII

First article inspection and tests

Inspection or Test	Dest	Non-Dest	Units To Be Tested 1-3	4-6	7-9
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X
2. *Non-actuation shock at room temp (see 3.6.1.1)		X	X		
3. Non-actuation shock at -65°F (see 3.6.1.1)		X		X	
4. Non-actuation shock at 160°F (see 3.6.1.1)		X			X
5. *No-load actuation shock at room temp (see 3.6.1.2)		X	X		
6. No-load actuation shock at -65°F (see 3.6.1.2)		X		X	
7. No-load actuation shock at 160°F (see 3.6.1.2)		X			X
8. Acceleration (see 3.6.2)		X	X		
9. *Vibration at room temp (see 3.6.3)		X	X		
10. Vibration at -65°F (see 3.6.3)		X		X	
11. Vibration at 160°F (see 3.6.3)		X			X
12. Temperature shock (see 3.6.4)	X		X		
13. Humidity (see 3.6.5)	X			X	
14. *50-ampere load actuation shock at room temperature (see 3.6.1.3)	X		X		
15. 50-ampere load actuation shock at -65°F (see 3.6.1.3)	X			X	
16. 50-ampere load actuation shock at 160°F (see 3.6.1.3)	X				X

*Room temperature = 80° ± 20°F.

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VIII and Table IX), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VIII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IX
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Impact fuze prior to potting (see dwg 8813373).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any part missing, loose or damaged	Visual
	102. Improper wire routing	Visual
	103. Improper soldering (see 3.7)	Visual
	104. Evidence of poor workmanship (see 3.8)	Visual
Minor:	None defined.	

4.3.2.1.2 Impact fuze after potting, (see dwg 8813373).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper location of mounting holes	Gage
	102. Improper distance between contacts	Gage
	103. Improper location of male contacts	Gage
	104. Wrong connector	Visual
	105. Deformation of connector sufficient to impair mating	Visual
	106. Deformation of contact	Visual
	107. Any pin missing	Visual

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	108.	Improper potting including wire protruding from potting.....Visual
	*109.	Height of pins (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins).....Visual
	110.	Objective evidence relative to material missing or improper (see 3.1).....Visual
	111.	Serialization improper or missing (see 3.4).....Visual
Minor:		
	201.	Evidence of poor workmanship (see 3.8).....Visual

*In the event pins do not appear to be within 1/16 inch in height relative to the adjacent pins, a gage (see 4.3.4.1) shall be used to determine that variation from the normal height. Reject any pin that varies by more than 0.063 inch maximum from the normal pin height of the individual connector. The normal height of the individual contact pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward edge of the connector shall be utilized as the reference surface for all measurements of pin heights.

4.3.2.1.3 Impact fuze packing (see dwg 8822043).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packing material visibly wet	Visual
	102. Incorrect or insufficient cushioning material	Visual
	103. Excess air in bag	Visual
	104. Marking missing, incorrect or unidentifiable	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.3.2.1.4 Packing, container (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Marking missing, incorrect or mislocated.....Visual
102. Incorrect or insufficient cushioning material.....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. *Non-actuation shock (at room temperature)	Major
b. *Actuation shock (at room temperature)	Major
c. Resistance	Major
d. Humidity	Major
e. Temperature shock	Major
f. *50-ampere load actuation shock (at room temperature)	Major

*Room temperature = 80° ± 20°F.

Of these tests, d, e and f are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Resistance, no-load actuation shock, and non-actuation shock. Each impact fuze shall be tested in accordance with 4.4.1, 4.4.2.1 and 4.4.2.2. If any item fails to comply with 3.5, 3.6.1.1 and 3.6.1.2, the item shall be rejected and removed from the lot.

4.3.3.2 Humidity, temperature shock, and 50-ampere load actuation shock. A sample shall be selected at random in accordance with Table X. The sample shall be tested in accordance with 4.4.3.4, 4.4.3.3 and 4.4.2.1. If any item fails to comply with 3.6.1.1, 3.6.4 and 3.6.5, the lot shall be rejected.

Table X

Lot Size	Sample Size
2-180	2
181-500	3
501-800	5
801-1300	7

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8842562). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Resistance.

4.4.1.1 Circuit resistance, actuated fuze. Through the use of a suitable fixture, depress the weight until contact 8813366 touches contact 8800854. Gradually remove force from the weight until the contacts separate. Add 1 pound ± 1 ounce to the force tending to hold the contacts together. This condition will simulate mechanical actuation of the fuze. The resistances shall then be checked as specified in Table I using a suitable resistance measuring device.

4.4.1.2 Circuit resistance, non-actuated condition. Using a potential of not more than 28 vdc and a suitable measuring device, the resistance shall be checked in accordance with Table II.

4.4.1.3 Insulation resistance. An insulation resistance tester, capable of supplying the required voltage and with which a resistance in megohms can be obtained, shall be used to measure the resistance. The test current shall be limited to a maximum of 5 milliamperes to protect human life. Requirements shall be as specified in 3.5.3.

4.4.2 Functional shock test.

4.4.2.1 Non-actuation shock. The fuze shall be mounted in a fixture which will permit shocks to be directed along three mutually perpendicular major axes. The shock testing

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apparatus shall be capable of subjecting the fuze in its fixture to a shock equivalent of a half-sine wave shock form with peak acceleration of 110 ± 10 g's and a time duration from beginning to end of such half-sine wave form of 10 ± 2 milliseconds. The above shock shall be applied one time along each of three mutually perpendicular major axes at an ambient temperature of 80° ± 20°F. During these shocks the circuit from AIJI Pin A to the cover shall be monitored using the circuit of Figure 1 or equivalent for evidence of closure. Prior to testing at -65° ± 5°F the impact fuze shall have been at -80 ± 5°F for a period of 4 hours minimum. Following this period the fuze shall be stabilized (see 6.4) at -65 ± 5°F for 1 hour minimum and then shock tested. The fuze shall then be stabilized at 160 ± 5°F and shock tested. Following the shock tests the fuze shall be tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.2.2 No-load actuation shock. The impact fuze shall be mounted in a fixture which will permit shocks to be directed along three mutually perpendicular major axes. The shock testing apparatus shall be capable of subjecting the fuze in its fixture to a shock equivalent to a half-size wave form shock with a peak acceleration of 250 ± 20 g's and a time duration from beginning to end of such half-sine wave form of 9 ± 3 milliseconds. The above shock shall be applied at an ambient temperature of 80 ± 20°F one time in each of three mutually perpendicular major axes. During these shocks the circuit between AIJI Pin A and the cover shall be monitored to determine the time duration of closure. Prior to testing at -65 ± 5°F, the impact fuze shall have been at -80 ± 5°F for a period of 4 hours minimum. Following this period the fuze shall be stabilized at -65 ± 5°F for 1 hour minimum and then shock tested while at this temperature. The fuze shall then be stabilized at 160 ± 10°F and shock tested while at this temperature. Following the shock test the fuze shall be tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.2.3 50-ampere load-actuation shock. Shock the fuze as in 4.4.2.2 and monitor the duration of closure and voltage drop from AIJI Pin A to the cover with a current load of 50 -5, ± 0 amperes. Each fuze shall be shocked in one axis only and one time only.

4.4.3 Environmental testing (see 3.6).

4.4.3.1 Acceleration. Subject the fuze to a steady acceleration for a period of at least 1 minute one time along each of three mutually perpendicular major axes. While the fuze is being accelerated, monitor the circuit from AIJI Pin A to the cover for evidence of closure (see Figure 1). This test shall be made at 80 ± 20°F. The assembly shall then be tested in accordance with 4.4.1.2 and 4.4.1.3.

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4.4.3.2 Vibration. The fuze shall be rigidly mounted to a fixture capable of being supported rigidly in each of three mutually perpendicular axes with the fixture mounted on a suitable vibration producing apparatus. The fuze shall be vibrated at the frequencies, amplitudes, temperatures, and for the time specified in Table VI in each of three mutually perpendicular major axes. Following this test the assembly shall be tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.3.3 Temperature shock. The fuze shall be placed within a test chamber where the temperature is -65 + 5°F. The fuze shall be subjected to this temperature for a period of 4 hours minimum at the conclusion of which, and within 5 minutes, the fuze shall be transferred to a chamber having an internal temperature of 160 + 5°F. The fuze shall be subjected to this temperature for a period of 4 hours minimum. This completes one cycle. The number of cycles shall be three with a maximum of 5 minutes transfer time between test chambers when going from one cycle to another. At the conclusion of the third cycle the fuze shall be removed from the test chamber. The fuze shall then be tested as specified in 4.4.1.2, 4.4.1.3, 4.4.2.1 and 4.4.2.2.

4.4.3.4 Humidity. The impact fuze shall be placed in the test chamber mounted to simulate the installed conditions. The pipe plug must be firmly in place on the top as specified in dwg 8813373. The chamber temperature shall be 80 + 20°F. During the first 2 hour period, the temperature shall be gradually raised to 160 + 5°F. The temperature of 160 + 5°F shall be maintained during the next 6 hour period. During the following 16 hour period, the temperature in the chamber shall be gradually reduced to 80 + 20°F which completes one cycle. The relative humidity throughout the cycle shall be 95 + 5 percent. The test shall consist of 15 such cycles (360 hours). The assembly shall then be removed from the test chamber and tested in accordance with 4.4.1.2, 4.4.1.3, 4.4.2.1 and 4.4.2.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The impact fuze shall be packaged in accordance with dwg 8822043 using the specified moisture barrier.

5.1.2 Level C. The impact fuze shall be packaged in accordance with dwg 8822043 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The impact fuze shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822043.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1680 dated 1 August 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A135

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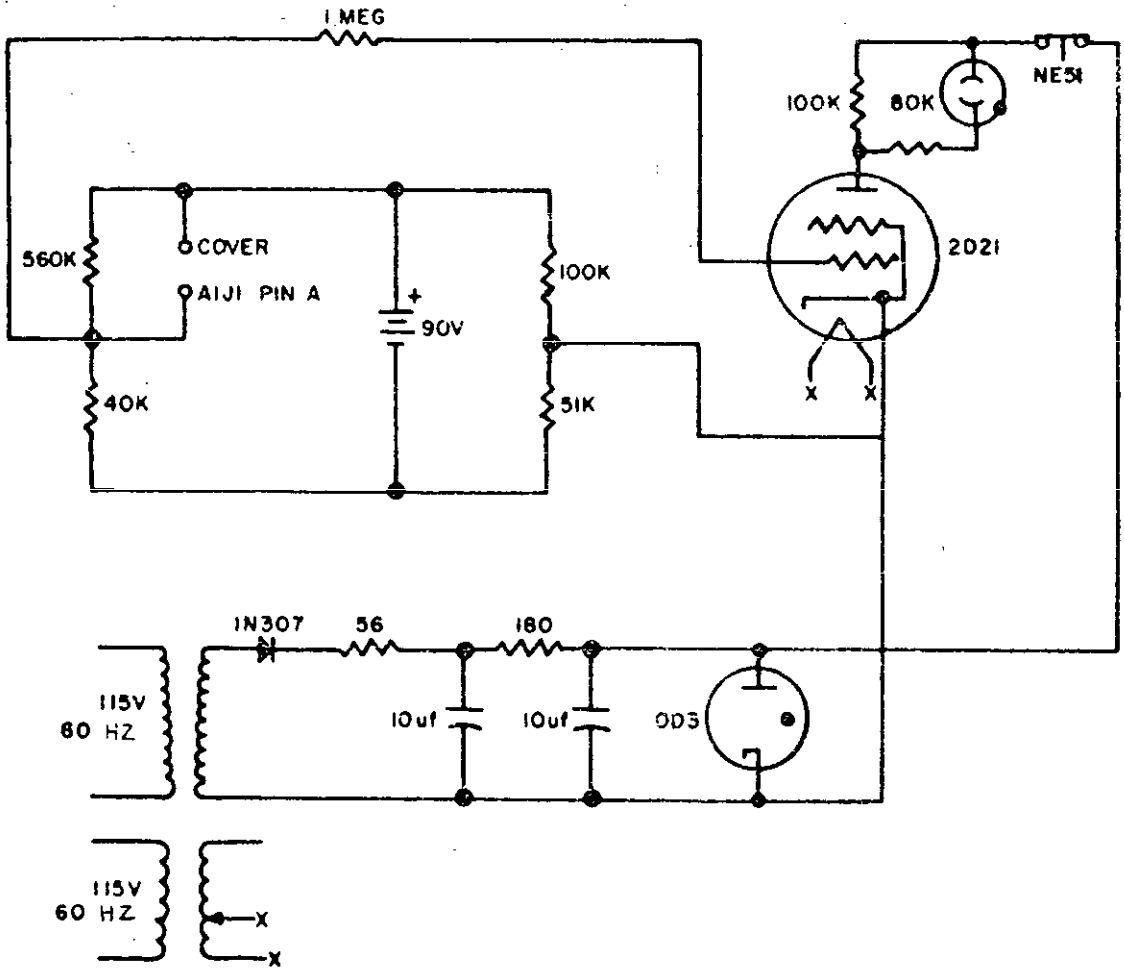


Figure 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

FUZE, IMPACT, ATOMIC WEAPON: M1110

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 — REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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